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Chemistry Paper 3A WASSCE (SC), 2018

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Question 1A

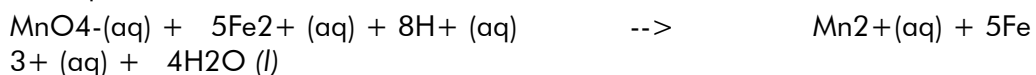
A is a solution of potassium tetraoxomanganate(VII).

B is a solution of iron(II)chloride containing 4.80 g of the salt in 250 cm³ of solution.

- Put A into the burette. Pipette 20.0 cm³ or 25.0 cm³ of B into a conical flask, add 20.0 cm³ of H₂SO₄ (aq) and titrate with A. Repeat the titration to obtain concordant titre value.

Tabulate your results and calculate the average volume of A used.

The equation of the reaction is:



(b) From your results and information provided, calculate the:

- concentration of B in mol dm⁻³;
 - concentration of A in mol dm⁻³;
 - number of mole of Fe²⁺ in the volume of B pipetted.
- [FeCl₂ = 127 g mol⁻¹][20 marks]

Observation

In part (a), majority of the candidates tabulated the titration table correctly and calculated the volume of A used.

In part (b), majority of the candidates calculated the concentrations of A and B in mol dm⁻³ correctly. Some candidates lost marks to wrong unit while majority of them did not understand the concept of mole ratio.

The expected answers include:

(a) Two concordant titres = 8 marks

Averaging = 2 marks

(b) (i) 250 cm³ of B contains 4.80 g of FeCl₂
 $\therefore 1000 \text{ cm}^3 \text{ will contain} = \frac{1000}{250} \times 4.80 \text{ g}$
 $= 19.2 \text{ g dm}^{-3}$

Alternative Method

$$\text{Mass conc. } \rho (\text{g dm}^{-3}) = \frac{\text{mass, } m (\text{g})}{\text{volume, } V (\text{dm}^3)}$$

Substituting $\rho = \frac{4.80 \text{ g}}{0.250 \text{ dm}^3}$

$$= 19.2 \text{ g dm}^{-3}$$

$$\begin{aligned} \text{Conc. of B in mol dm}^{-3} &= \frac{\text{Conc. in g dm}^{-3}}{\text{Molar mass}} \\ &= \frac{19.2 \text{ g dm}^{-3}}{127 \text{ g mol}^{-1}} \\ &= 0.151 \text{ mol dm}^{-3} \end{aligned}$$

Alternative Method

$$\text{Conc, } C, (\text{mol dm}^{-3}) = \frac{\text{amount, } n (\text{mol})}{\text{volume} (\text{dm}^3)}$$

$$= \frac{\text{mass, } m (\text{g})}{\text{molar mass, } M (\text{g mol}^{-1}) \times V (\text{dm}^3)}$$

i.e. $C = \frac{m}{MV}$

Substituting $C = \frac{4.80}{127 \times 0.250}$

$$= 0.151 \text{ mol dm}^{-3}$$

(ii) $\frac{C_A V_A}{C_B V_B} = \frac{1}{5}$ -----mole ratio

$C_A = \frac{C_B V_B n_A}{V_A \times n_B}$ ----- making C_A subject of formula

$= \frac{0.151 \times 25.0 / 20.0}{V_A \times 5}$ ----- correct substitution

= Say amol dm^{-3} ----- correct evaluate to 3 sig fig

Alternative method

Amount of B used = $0.151 \times \frac{V_B}{1000}$

From the balanced equation of reaction

5 moles of B $\equiv$ 1 mole of A

Hence, amount of A = $\frac{0.151 \times V_B}{5 \times 1000}$

i.e. $V_A \text{ cm}^3$ contained $\frac{0.151 \times V_B}{5 \times 1000}$

1000 cm^3 contained, $\frac{0.151 \times V_B}{5 \times 1000} \times \frac{1000}{V_A}$

= $\frac{0.151 \times V_B}{5}$

$$V_A = \text{Say } a \text{ mol dm}^{-3}$$

- (iii) 1000 cm³ of B contains 0.151 moles of Fe²⁺
 $\therefore 20.0 / 25.0 \text{ cm}^3$ will contain, $\frac{0.151 \times 25.0 / 20.0}{1000}$
 Say **b mol**

Alternative method

$$\text{Amount of B} = \frac{0.151 \times V_B}{5 \times 1000}$$

$$= \text{Say } b \text{ mol}$$

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